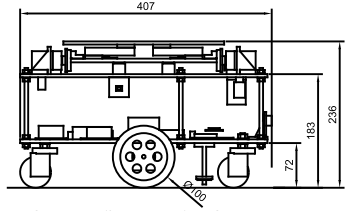
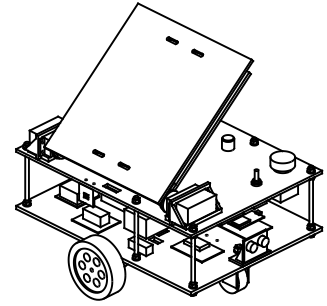


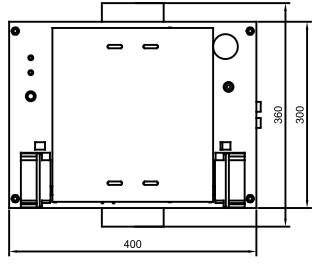
FRONT VIEW



SIDE VIEW (from the right) FRONT -->



ISOMETRIC VIEW - BED TIPPED 40 deg

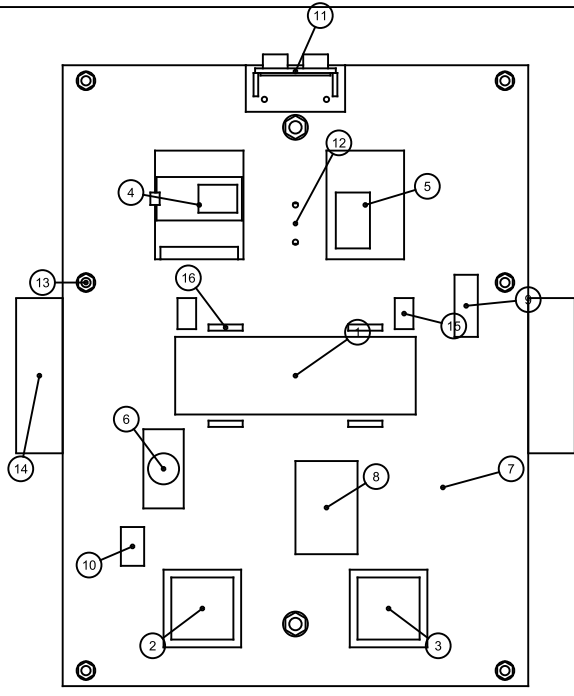


PLAN VIEW FRONT -->

NOTES

1. All dimensions in mm. Weights and loads in kg.
2. Drive: 2 x JGB37-520 12 V 170 RPM gear motors, wheels D100 x 30.
3. Casters: 2 x 50 mm swivel, M8 threaded stem (height set by nuts).
4. Frame: 6 mm plywood + 5 mm acrylic on 6 x M6 threaded rod.
5. Bed: tipped 40 deg to the right by 2 servos on the hinge line.
6. Payload: 4 x 10 kg load cells under the cargo plate; packages up to 3 kg.
7. ESP32 + 8-ch IR line sensor (under), HC-SR04 (front), VL53L0X (right side).
8. Sheets: 01 GA, 02 layout, 03-06 plates, 07 bed, 08-09 prints, 10 BOM.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: GENERAL ARRANGEMENT					DWG No.: CR-V2-001
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:5	SHEET: 1 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 22/10/2026	DRAWN:



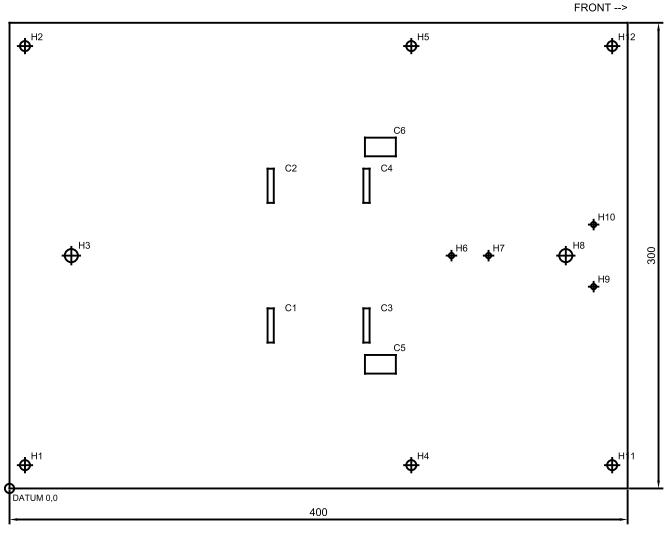
PLAN VIEW - DECK AND BED REMOVED (front of robot at the top)

No.	COMPONENT
1	3S LiPo battery 5000 mAh (lies across, slides out sideways)
2	BTS7960 motor driver - left motor
3	BTS7960 motor driver - right motor
4	ESP32 DevKit on a screw-terminal breakout (USB to the left)
5	I/O board: MCP23017 + resistor dividers
6	Buck converter 12 V -> 5.1 V (ESP32, sensors)
7	Buck converter 12 V -> 7.4 V (bed servos)
8	WAGO power buses (+12 V main / motor / 5 V / GND)
9	Inline fuse 10 A
10	Buzzer
11	HC-SR04 ultrasonic on printed bracket
12	8-channel IR line sensor (under the plate)
13	M6 threaded rod (x6)
14	Drive wheel D100 (motor under the plate)
15	Motor / encoder cable slot (x2)
16	Battery strap slots (x4)

NOTES

Small boards are held with VHB tape.
Keep 10 cm slack on every wire that goes up to the deck or the bed.
Wiring: see BUILD_GUIDE.md weeks 2 and 3, and the schematics E01-E04.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: LAYOUT - DECK REMOVED				DWG No.: CR-V2-002	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 2 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 22/10/2026	DRAWN:



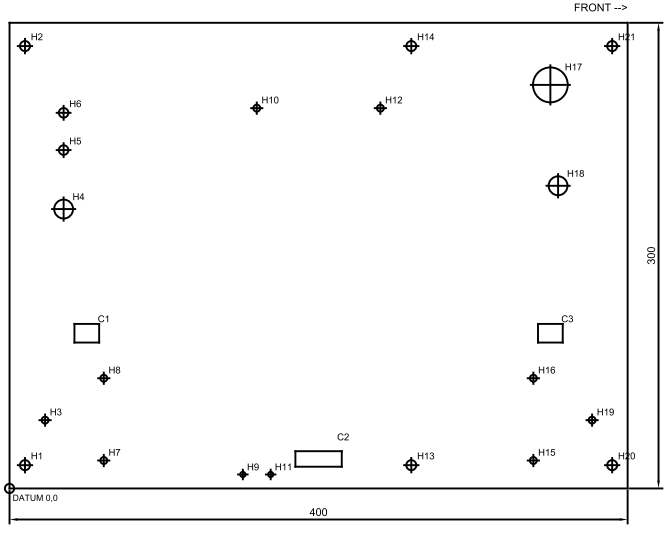
ID	X	Y	SIZE	FOR
H1	10.0	15.0	D6.5	M6 threaded rod
H2	10.0	285.0	D6.5	M6 threaded rod
H3	40.0	150.0	D8.5	caster stem M8
H4	260.0	15.0	D6.5	M6 threaded rod
H5	260.0	285.0	D6.5	M6 threaded rod
H6	286.0	150.0	D3.4	M3 sensor brackets
H7	310.0	150.0	D3.4	M3 sensor brackets
H8	360.0	150.0	D8.5	caster stem M8
H9	378.0	130.0	D3.4	M3 sensor brackets
H10	378.0	170.0	D3.4	M3 sensor brackets
H11	390.0	15.0	D6.5	M6 threaded rod
H12	390.0	285.0	D6.5	M6 threaded rod
C1	169.0	105.0	4 x 22	battery strap
C2	169.0	195.0	4 x 22	battery strap
C3	231.0	105.0	4 x 22	battery strap
C4	231.0	195.0	4 x 22	battery strap
C5	240.0	80.0	20 x 12	motor/encoder cables
C6	240.0	220.0	20 x 12	motor/encoder cables

X, Y = centre of the feature from DATUM 0,0 (H = hole, S = slot, C = cut-out)

NOTES

1. Material: 6 mm birch plywood (or 5 mm aluminium), 1 off.
2. Laser-cut from cad/output/dxf/bottom_plate.dxf. All dimensions in mm, +-0.2.
3. Motor-bracket holes (4 x D4.5) are drilled on assembly - see BUILD_GUIDE week 1.
4. Small holes: 2 for the line-sensor bracket (underneath), 2 for the ultrasonic bracket.

PROJECT					
AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE					DWG No.
BOTTOM PLATE					CR-V2-003
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE	SHEET	UNITS	PROJECTION	DATE	DRAWN
1:2	3 / 10	mm	1st ANGLE	22/10/2026	



ID	X	Y	SIZE	FOR
H1	10.0	15.0	D6.5	M6 threaded rod
H2	10.0	285.0	D6.5	M6 threaded rod
H3	23.0	44.0	D4.5	M4: servo mount / bed rest
H4	35.0	180.0	D12.2	switch / start button
H5	35.0	218.0	D6.2	5 mm LED clip
H6	35.0	242.0	D6.2	5 mm LED clip
H7	61.0	18.0	D4.5	M4: servo mount / bed rest
H8	61.0	71.0	D4.5	M4: servo mount / bed rest
H9	151.0	9.0	D3.4	M3: side sensor
H10	160.0	245.0	D4.5	M4: servo mount / bed rest
H11	169.0	9.0	D3.4	M3: side sensor
H12	240.0	245.0	D4.5	M4: servo mount / bed rest
H13	260.0	15.0	D6.5	M6 threaded rod
H14	260.0	285.0	D6.5	M6 threaded rod
H15	339.0	18.0	D4.5	M4: servo mount / bed rest
H16	339.0	71.0	D4.5	M4: servo mount / bed rest
H17	350.0	260.0	D22.5	E-stop 22 mm
H18	355.0	195.0	D12.2	switch / start button
H19	377.0	44.0	D4.5	M4: servo mount / bed rest
H20	390.0	15.0	D6.5	M6 threaded rod
H21	390.0	285.0	D6.5	M6 threaded rod
C1	50.0	100.0	16 x 12	servo cable
C2	200.0	19.0	30 x 10	load-cell wires
C3	350.0	100.0	16 x 12	servo cable

X, Y = centre of the feature from DATUM 0,0 (H = hole, S = slot, C = cut-out)

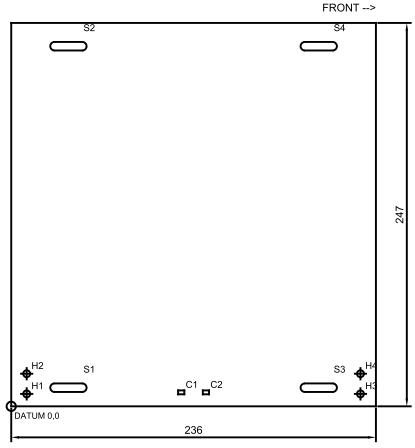
NOTES

1. Material: 5 mm clear acrylic (keep the protective film), 1 off.
2. Laser-cut from cad/output/dxf/top_deck.dxf. All dimensions in mm, +-0.2.
3. Every hole is cut by the laser; nothing is drilled by hand.

PROJECT					
AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE					DWG No.
TOP DECK					CR-V2-004
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE	SHEET	UNITS	PROJECTION	DATE	DRAWN
1:2	4 / 10	mm	1st ANGLE	22/10/2026	

ID	X	Y	SIZE	FOR
H1	10.0	8.0	D4.5	M4 horn adapter
H2	10.0	21.0	D4.5	M4 horn adapter
H3	226.0	8.0	D4.5	M4 horn adapter
H4	226.0	21.0	D4.5	M4 horn adapter
S1	37.0	12.0	23.5 x 5.5 (R2.75)	load cells, fixed ends
S2	37.0	232.0	23.5 x 5.5 (R2.75)	load cells, fixed ends
S3	199.0	12.0	23.5 x 5.5 (R2.75)	load cells, fixed ends
S4	199.0	232.0	23.5 x 5.5 (R2.75)	load cells, fixed ends
C1	110.0	9.0	4 x 2.5	cable tie
C2	126.0	9.0	4 x 2.5	cable tie

X, Y = centre of the feature from DATUM 0,0 (H = hole, S = slot, C = cut-out)



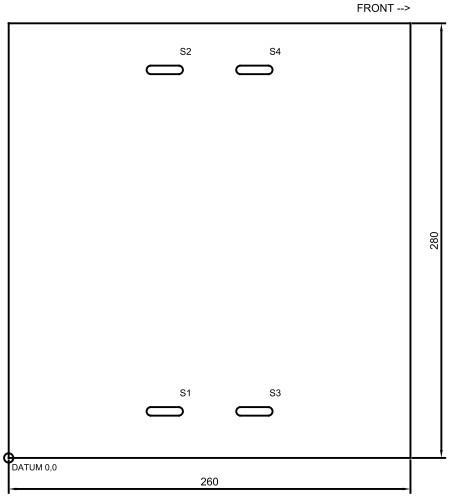
NOTES

1. Material: 6 mm birch plywood, 1 off.
2. Laser-cut from cad/output/dxf/tilt_plate.dxf. All dimensions in mm, +-0.2.
3. The bottom edge here (Y = 0) is the RIGHT edge of the robot: it lies on the hinge axis.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: TILTING PLATE					DWG No.: CR-V2-005
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 5 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 22/10/2026	DRAWN:

ID	X	Y	SIZE	FOR
S1	101.0	30.0	23.5 x 5.5 (R2.75)	load cells, loaded ends
S2	101.0	250.0	23.5 x 5.5 (R2.75)	load cells, loaded ends
S3	159.0	30.0	23.5 x 5.5 (R2.75)	load cells, loaded ends
S4	159.0	250.0	23.5 x 5.5 (R2.75)	load cells, loaded ends

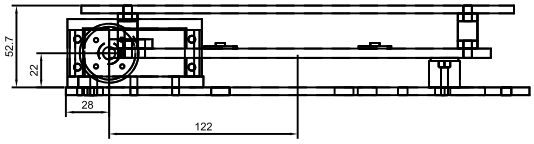
X, Y = centre of the feature from DATUM 0,0 (H = hole, S = slot, C = cut-out)



NOTES

1. Material: 5 mm clear acrylic, 1 off. No anti-slip mat: the package must slide off.
2. Laser-cut from cad/output/dxf/cargo_plate.dxf. All dimensions in mm, +-0.2.
3. Rests only on the 4 load cells (slots S1-S4).

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: CARGO PLATE					DWG No.: CR-V2-006
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 6 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 22/10/2026	DRAWN:



BED FLAT - FROM THE FRONT (front servo not shown)

HOW IT WORKS

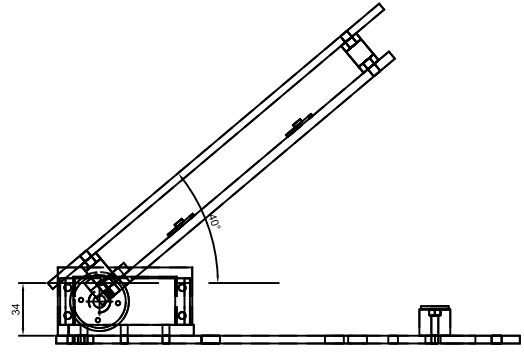
1. Two servos on the deck's right edge share one axis: their shafts ARE the hinge (front and rear end of the bed, mirror images). No hinge, screw or lever.
2. A printed horn adapter on each end of the tilting plate screws to the servo's aluminium disc horn (4 x M3). The load cells ride on the tilting plate.
3. Flat, the bed rests on 2 printed posts and the servos are switched off.
4. To unload, the ESP32 turns both servos slowly (20 deg/s) to 40 deg, waits until the load cells read empty, then lowers the bed and switches them off.
5. No anti-slip mat: a cardboard box slides on acrylic from about 20 deg.

TORQUE AT THE START OF THE TIP (both servos, 20 cm box)

PACKAGE	TORQUE	OF 2 x 60 kg.cm
1 kg	2.3 N.m	20 %
3 kg	4.7 N.m	40 %
5 kg	7.1 N.m	60 %

SERVOS

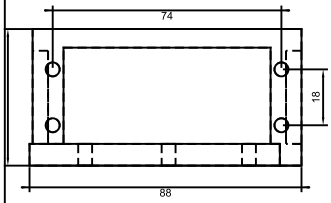
Servo: 60 kg.cm at 7.4 V, metal gears, 180 deg, 25T spline,
about 65 x 30 x 48 mm (e.g. DS5160). Check its tab holes:
74 x 18 mm here (SERVO in robot_cad.py).
Supply: 7.4 V from its own 10 A buck converter, after the E-stop.
Signal: front servo GPIO 2, rear servo GPIO 12 (schematic E03).
Hinge axis: 205 mm above the floor, 122 mm right of the centre line.



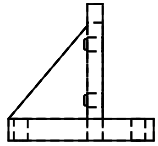
BED TIPPED 40 deg - THE PACKAGE SLIDES OFF TO THE RIGHT

(the right edge of the cargo plate ends up 6 mm outside the deck, 34 mm above it)

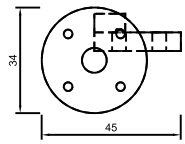
PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: TILTING BED - SERVO HINGE				DWG No.: CR-V2-007	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 7 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 22/10/2026	DRAWN:

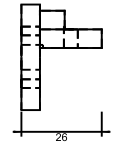


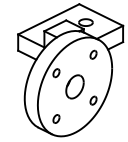
SERVO MOUNT (front) - FROM THE BED SIDE



SIDE

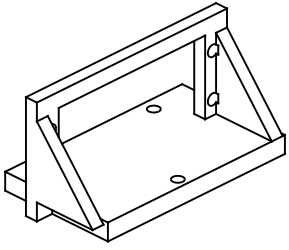




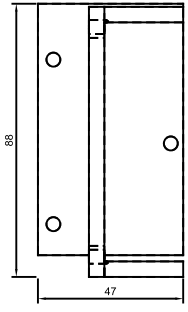


HORN ADAPTER (front) - FROM THE SERVO

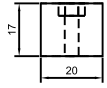
4 x D2.8 on a D24 circle: the horn's M3 screws cut their own thread.
2 x D4.5: M4 bolts down through the tilting plate. Rear = mirror.



4 x D4.5: servo tabs (M4 + nut).
3 x D4.5: bolts to the deck. Rear mount = mirror image.

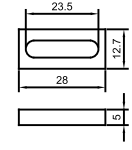


PLAN



BED REST (2 off)

D4.5 through; M4 nut pocket at the top, under the 2 mm rubber pad

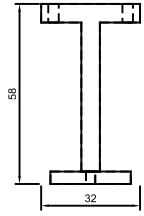


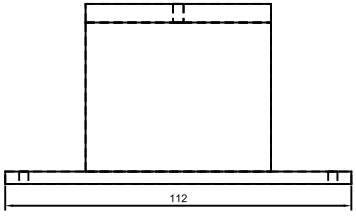
LOAD-CELL SPACER (8 off), slot 5.5 wide

NOTES

1. Material PETG, 0.2 mm layers, 4 walls. STL files: cad/output/parts/
2. Servo mounts: 1 + 1 (front + rear), 40 % infill, foot flat on the bed.
3. Horn adapters: 1 + 1, 100 % infill, round wall flat on the bed.
4. Bed rests: 2 off, 40 % infill. Spacers: 8 off, 100 % infill, flat.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: PRINTED PARTS (1/2) - BED					DWG No.: CR-V2-008
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:1	SHEET: 8 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 22/10/2026	DRAWN:

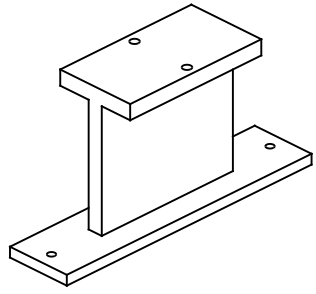


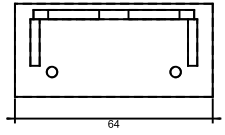


LINE SENSOR BRACKET - SIDE FROM THE FRONT

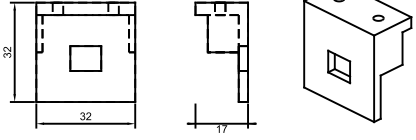
Flange under the bottom plate: 2 x D3.4 on the centre line, 24 mm apart (X = 98 +-12).

The sensor board hangs under the shelf: 2 x D3.0 (M3), 100 mm apart.



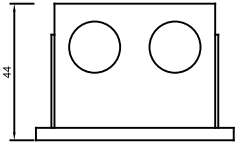


ULTRASONIC BRACKET - PLAN (front up)



SIDE SENSOR BRACKET (VL53L0X)

Hangs under the deck (2 x M3); 10 x 8 window for the sensor.



FROM THE FRONT - windows D16.5 for the transducers

NOTES

1. Material PETG, 0.2 mm layers, 4 walls, 40 % infill. STL: cad/output/parts/
2. Line sensor bracket: flange flat on the bed, supports under the shelf.
3. Ultrasonic bracket: base flat on the bed, no supports.
4. Side sensor bracket: top flange flat on the bed, no supports.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: PRINTED PARTS (2/2) - SENSORS				DWG No.: CR-V2-009	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:1	SHEET: 9 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 22/10/2026	DRAWN:

No.	GROUP	DESCRIPTION	QTY	MAKE / BUY
1	Compute	ESP32 DevKitC 38-pin (ESP32-WROOM-32, USB) + 38-pin screw-terminal breakout board	1	buy (get 2)
2	Compute	MCP23017 I2C 16-bit I/O expander module (address 0x20)	1	buy
3	Drive	JGB37-520 12V 170 RPM gear motor with Hall encoder (6 wires)	2	buy
4	Drive	Steel mounting bracket for 37 mm gear motor (+ its M3 screws)	2	buy
5	Drive	Wheel 100 mm x 30 mm with 6 mm D-shaft hub	2	buy
6	Drive	Swivel caster 50 mm, M8 threaded stem, >= 20 kg rating	2	buy
7	Drive	BTS7960 (IBT-2) 43 A motor driver	2	buy
8	Sensors	8-channel IR line sensor array, TCRT5000, digital outputs, 3.3-5 V	1	buy
9	Sensors	Line sensor bracket (line_sensor_bracket.stl)	1	3D print PETG
10	Sensors	HC-SR04 ultrasonic distance sensor	1	buy (get 2)
11	Sensors	Ultrasonic sensor bracket (sonar_bracket.stl)	1	3D print PETG
12	Sensors	VL53L0X time-of-flight distance sensor module	1	buy
13	Sensors	Side sensor bracket (tof_bracket.stl)	1	3D print PETG
14	Payload	Straight-bar load cell 10 kg, 80x12.7x12.7 mm	4	buy
15	Payload	HX711 module	4	buy
16	Payload	Load-cell spacer (load_cell_spacer.stl)	8	3D print PETG 100 %
17	Payload	Cargo plate 260x280x5 (cargo_plate.dxf)	1	laser-cut acrylic
18	Tilting bed	Tilting plate 236x247x6 (tilt_plate.dxf)	1	laser-cut 6 mm plywood
19	Tilting bed	Servo 60 kg.cm, metal gears, 180 deg, 6-8.4 V, 25T spline, about 65x30x48 mm (e.g. DS5160)	2	buy
20	Tilting bed	25T aluminium disc servo horn, about 38 mm (often included with the servo)	2	buy
21	Tilting bed	Servo mount (servo_mount_front.stl + servo_mount_rear.stl)	2	3D print PETG
22	Tilting bed	Horn adapter (horn_adapter_front.stl + horn_adapter_rear.stl)	2	3D print PETG 100 %
23	Tilting bed	Bed rest post (bed_rest.stl) + 18 mm self-adhesive rubber pad 2 mm	2	3D print PETG
24	Tilting bed	Buck converter 10 A (e.g. 300 W XL4016 module), set to 7.4 V for the servos	1	buy
25	Power	3S LiPo 11.1 V 5000 mAh XT60 + balance charger + LiPo bag + voltage alarm	1	buy
26	Power	XL4015 5 A buck converter (5.1 V for the ESP32 and sensors)	1	buy
27	Power	Inline fuse holder + 10 A fuses, WAGO 221 connectors, XT60 pairs	1	buy
28	Power	Emergency stop 22 mm panel-mount, latching, NC contact block	1	buy
29	Power	Toggle switch 12 mm panel-mount, 12 V DC >= 15 A	1	buy
30	Controls	Push button 12 mm panel-mount, momentary, normally open (start)	1	buy
31	Controls	5 mm LED green + red, with panel clips, + 330 ohm resistors	1	buy
32	Controls	Active buzzer module, 3-pin, 3.3-5 V	1	buy
33	Chassis	Bottom plate 400x300x6 (bottom_plate.dxf)	1	laser-cut 6 mm plywood
34	Chassis	Top deck 400x300x5 (top_deck.dxf)	1	laser-cut 5 mm acrylic
35	Chassis	M6 threaded rod, cut to 130 mm	6	cut from 1 m rod
36	Chassis	M6 nut + M6 washer	24	buy
37	Chassis	M8 nut + M8 washer (casters)	4	buy
38	Hardware	M4x12 screws + nylock nuts (motor brackets, servo mounts, rest posts), M4x16 + nuts (horn adapters), M4/M5x16 (load cells), M3 screws + nuts (sensor brackets), M2.5/M3 for the boards, VHB tape, velcro strap, zip ties	1	buy
39	Hardware	Resistors 1k x1, 2k x1, 10k x5, 2.2k x1, 4.7k x2, 100 nF x1, small perfboard x2	1	buy

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE: BILL OF MATERIALS					DWG No.: CR-V2-010
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: -	SHEET: 10 / 10	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 22/10/2026	DRAWN: